

A PROJECT REPORT ON

**Smart City Exploration: A Review-Based
Recommendation System**

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SUBMITTED BY

Jatin Yadav
Sushant Langhi
Kartik Kirve
Ayush Meshram

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Exam No: B400050439
Exam No: B400050504



**DEPARTMENT OF COMPUTER ENGINEERING
Pune Institute of Computer Technology
Dhankawadi, Pune - 411043**

**SAVITRIBAI PHULE PUNE UNIVERSITY
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CERTIFICATE

This is to certify that the project report entitled
Smart City Exploration: A Review-Based Recommendation System

Submitted by

Jatin Yadav
Sushant Langhi
Kartik Kirve
Ayush Meshram

Exam No: B400050424
Exam No: B400050472
Exam No: B400050439
Exam No: B400050504

is a bonafide student of this institute and the work has been carried out by him/her under the supervision of **Asst. Prof. P. T. Kohok** and it is approved for the partial fulfillment of the requirement of Savitribai Phule Pune University, for the award of the degree of **Bachelor of Engineering** (Computer Engineering).

Asst. Prof. P. T. Kohok
Internal Guide
Dept. of Computer Engg.

Dr. G. V. Kale
Head
Dept. of Computer Engg.

Dr. S. T. Gandhe
Principal
Pune Institute of Computer Technology

Place: Pune
Date:

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Jatin Yadav
Sushant Langhi
Kartik Kirve
Ayush Meshram
(B.E. Computer Engg.)

ABSTRACT

In the fast-paced world of tourism and travel, personalized recommendations are becoming a crucial tool to assist individuals in making well-informed choices about destinations, events, and places to visit. Our project, Smart City Exploration: A Review-Based Recommendation System, aims to deliver tailored suggestions based on user preferences, utilizing data sourced from APIs like Google Maps and leveraging Natural Language Processing (NLP) techniques to enhance the relevance and accuracy of the recommendations.

With a vast amount of data available on the web, travelers today have an overwhelming number of options to choose from when planning trips. While existing platforms offer some level of recommendations, they often lack the depth of customization needed to match individual user preferences, especially when factoring in personal preferences such as travel style, seasonality, and time of visit. The rise of review-driven platforms means users expect not only factual data about places, but also insightful sentiment-based suggestions tailored to their needs.

Our system bridges the gap between static travel suggestions and personalized experiences by incorporating user reviews, sentiment analysis, and preference-based matching. This offers users a more accurate, relevant, and meaningful set of travel recommendations. By continuously learning from user feedback and real-time data, the system will evolve to provide even more precise suggestions, creating a future-proof solution for modern travelers.

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CHAPTER 1

INTRODUCTION

1.1 Overview

A User Review-Based Place Recommendation System is a sensible answer that leverages person-generated critiques to indicate places of hobby in smart town surroundings. Our recommendation system is predicated on crowd-sourced data from people who've visited numerous places, imparting their experiences, reviews, and scores. The intention is to offer personalized guidelines to users based totally on the aggregated insights from those reviews, helping them discover places that align with their options. As towns become smarter and more connected, such structures play a critical role in improving consumer reviews by means of supplying curated guidelines based on real-life comments.

1.2 Motivation

This project stems from the growing need for reliable and personalized recommendations for places to visit, especially in a world where information is abundant but often overwhelming. With the rapid increase in user-generated content, such as reviews and ratings on platforms like Google Maps, it has become increasingly difficult for users to determine the true quality and relevance of a location. At the same time, the presence of fake or misleading reviews has compromised the integrity of these platforms.

This project addresses these challenges by using machine learning techniques to extract meaningful insights from reviews and metadata, providing users with highly accurate, personalized recommendations. By categorizing places based on keywords, review content, and metadata, the system can suggest locations that align closely with a user's preferences—whether PICT, Department of Computer Engineering 2024-25 12 for a family outing, adventure, or solo exploration. This approach improves the overall user experience by filtering out unreliable information and focusing on authentic feedback, thus empowering users to make informed decisions.

1.3 Problem Definition and Objectives

A man visiting a foreign city with his family struggles to find suitable places using Google Maps due to overwhelming information and inaccurate reviews. Google Maps provides ratings and basic details but lacks personalized filters for preferences like family-friendliness, budget, or accessibility. This project

addresses these issues by developing a recommendation system that analyzes real-time data from platforms like Google Maps. It filters out fake reviews using machine learning, analyzes metadata, and delivers personalized recommendations based on factors such as family-friendliness, accessibility, and budget, ensuring a seamless and enjoyable experience for users.

1.4 Project Scope & Limitations

The travel manual recommendation system will leverage Google Maps API, consumer-generated possibilities, and Natural Language Processing (NLP) to curate customized recommendations for places to go to. The system will endorse places primarily based on user possibilities along with pastime type (solo, own family, date), season, crowd tiers, and accessibility. The system will function as a frontend built using React.js, making sure a responsive consumer interface, and a backend powered by using Node.js for coping with API requests and coping with statistics. Python may be hired for the NLP responsibilities, consisting of sentiment evaluation of opinions to refine suggestions.

1.5 Methodologies of Problem solving

The methodology follows a systematic approach to process and analyze raw data, detect fake reviews, and generate accurate recommendations. The entire workflow involves several stages, starting from data collection and cleaning to implementation the final recommendation system.

A. Data Collection and Cleaning

The dataset used in this study was compiled through a combination of the Google Maps API and web scraping techniques, and was stored in Excel format. It includes key information such as place names, business categories, user reviews, ratings, and other associated metadata. The initial step involved identifying and handling missing, incomplete, or irrelevant data to improve data quality and ensure consistency.

B. Data Filtering

Following the data cleaning phase, the dataset is further refined by applying filters that prioritize reliability and user relevance. One of the key filters

involves retaining only those businesses or places that have received a substantial number of user reviews—typically, entries with 100 or more reviews. This threshold ensures that the data used for recommendations is backed by a sufficient volume of user feedback, thereby reducing the risk of bias from sparse or unrepresentative entries.

C. Text Preprocessing and Word2Vec Embeddings

The textual data, primarily composed of user reviews, undergoes several preprocessing steps to prepare it for machine learning tasks. First, all text is converted to lowercase to ensure uniformity, and non-alphabetic characters such as punctuation marks and numbers are removed. Tokenization is then applied, breaking each review into individual words. Common stopwords (e.g., “and”, “the”, “is”) are eliminated as they do not provide meaningful insights

D. Classification using Machine Learning

To classify the businesses based on their type (such as restaurant, hotel, etc.), the categorical label type is encoded into numeric values using label encoding. The dataset is then split into training and testing subsets, ensuring the model can be evaluated fairly. A Random Forest classifier is trained on the processed review embeddings. This model is chosen because of its robustness and ability to handle complex, high-dimensional datasets. The classifier’s performance is evaluated using accuracy, and fine-tuned for better prediction results.

E. Sentiment Analysis

Sentiment analysis is performed to understand the overall user perception of a business. Each review’s sentiment score is computed using the TextBlob library, which provides a polarity score indicating whether a review is positive, negative, or neutral. This sentiment information can be used to gauge the general public’s opinion of a business, providing insights that can enhance the recommendation system.

F. Fake Review Detection

Detecting fake reviews is a critical component of ensuring the quality of recommendations. Fake reviews are often characterized by extreme sentiment scores, excessive repetition, or unusually short/long lengths

G. Data Transformation and Expansion

In this stage, nested data, such as JSON formatted columns, are expanded and transformed into a tabular format. The JSON data is parsed, flattened, and merged with the main dataset to create a comprehensive table. Null values in the JSON fields are replaced with a default value (False) to maintain consistency across the dataset. This transformed data, containing both textual reviews and structured metadata, is ready for integration with the machine learning models and the recommendation system.

H. Database Integration

Once the data is cleaned, transformed, and enriched with additional features, it is stored in a MySQL database for easier access and querying. Using Python's SQLAlchemy, the cleaned DataFrame is inserted into a MySQL table, where the table schema is dynamically generated to match the structure of the DataFrame. Special attention is given to handling missing values by converting them to SQL NULL.

I. System Design

The final recommendation system is deployed as a Flask-based web application. User authentication features, including Google OAuth, are integrated to allow secure access to the system. The recommendation system leverages the processed data to suggest relevant places to users based on their preferences. The application also integrates Leaflet.js for interactive map visualization, allowing users to view recommendations on a map and filter them according to different criteria.

CHAPTER 2

LITERATURE SURVEY

I. A significant focus in recommendation systems has been on understanding user preferences and tailoring suggestions accordingly. For example, Koren et al. (2009) explored collaborative filtering techniques, which analyze user behavior and preferences to make recommendations. These approaches have been enhanced by integrating user demographics and contextual information, as highlighted by Gonzalez et al. (2017) in their review of context-aware recommendation systems.

II. The analysis of user reviews plays a crucial role in enhancing recommendation accuracy. Techniques like Natural Language Processing (NLP) and sentiment analysis are employed to filter out fake reviews and assess the quality of places. Liu et al. (2015) provided an overview of sentiment analysis methods that can identify and categorize sentiments expressed in reviews, enabling the extraction of valuable insights from user-generated content. Similarly, Zhang et al. (2019) applied machine learning algorithms to classify reviews based on sentiment, contributing to more reliable recommendations.

III. Machine learning algorithms are instrumental in processing large datasets to identify patterns and make predictions. Rashidi et al. (2020) demonstrated the use of K-means clustering to categorize travel destinations based on user reviews and metadata, enhancing the classification of places into relevant categories. Additionally, Xiao et al. (2021) discussed the application of ensemble learning techniques that combine multiple models to improve recommendation accuracy, addressing issues related to data sparsity and cold starts.

IV. The cold-start problem—where a system struggles to provide recommendations for new users or items—remains a challenge in recommendation systems. Techniques to mitigate this include hybrid approaches that combine collaborative and content-based filtering. Coffey et al. (2019) highlighted strategies for enhancing initial recommendations by incorporating demographic data and leveraging social network information to bridge the gap until sufficient user data is collected.

V. Personalization remains a key focus in recommendation systems. Panniello et al. (2009) discussed how personalization techniques, such as user profiling and preference elicitation, can significantly enhance user satisfaction. By understanding the unique preferences of users, systems can recommend places that align closely with their interests.

VI. Despite advancements, several challenges remain in implementing effective recommendation systems. Issues such as data privacy, scalability, and maintaining the quality of recommendations are ongoing concerns. Ricci et al. (2015) outlined challenges in recommender systems, emphasizing the need for robust frameworks to ensure user trust and satisfaction

CHAPTER 3
SOFTWARE REQUIREMENTS
SPECIFICATION

3.1 Assumptions and Dependencies

In this section, we study the Assumptions and Dependencies of our project, user classes and characteristics along with assumptions and dependencies of our project.

3.2 Functional Requirements

3.2.1 System Feature 1(Functional Requirement)

- Input: User creates an account and inputs preferences such as preferred travel type, season, and budget.
- Process: Preferences are stored in a database (e.g., MongoDB) and matched with place metadata and review analysis data.
- Output: The system provides customized recommendations that match the user's preferences, with details on place type, open hours, and crowd levels.

3.2.2 System Feature2 (Functional Requirement)

- Input: The system receives user preferences and processes data from Google Maps API (place hours, reviews, etc.).
- Process: The engine applies algorithms that match user preferences with place tags (e.g., cultural, family-friendly). Sentiment analysis of reviews adjusts scores accordingly
- Output: The user is provided with a list of ranked places based on relevance to their preferences, including scores for categories like family-friendliness and accessibility.

3.3 External Interface Requirements

3.3.1 User Interfaces

The user interface is designed to be simple and intuitive, allowing users to interact with the recommendation system seamlessly. It provides input forms for users to specify their preferences, such as type of places, location, rating range, and more. Additionally, it includes an interactive map for users to view the recommended places, along with filters to customize the results.

3.3.2 Hardware Interfaces

- Server: The recommendation system runs on a cloud-based server, ensuring scalability and availability.
- User Devices: The system is accessible via web browsers on devices like laptops, desktops, and smartphones.

3.3.3 Software Interfaces

1. Operating System: Anyone among MAC OSX, Linux or Windows
2. Jupyter Notebook
3. Dataset pertinent to location-based reviews and place information
4. Programming Language: Python
5. Web Framework: Flask (for web deployment)
6. Visualization: Leaflet.js (for interactive maps)
7. Libraries: pandas, scikit-learn, TextBlob (for machine learning and sentiment analysis)

3.3.4 Communication Interfaces

The system uses HTTP/HTTPS protocols for communication between the web application and the backend server. Data exchanges between the client-side interface and the server are done using RESTful APIs. Additionally, Google OAuth is used for secure user authentication, ensuring user data privacy and security during interaction with the system.

1. Operating System: Anyone among MAC OSX, Linux or Windows

2. Jupyter Notebook
3. Dataset pertinent to medical images
4. Programming Language: Python

3.4 Nonfunctional Requirements

3.4.1 Performance Requirements

- **Scalability:** The system should support up to 10,000 concurrent users.
- **Latency:** API calls should be handled within 2 seconds, with NLP processing of reviews happening asynchronously to minimize delays.
- **Data Updates:** Data fetched from Google Maps API should be updated at regular intervals (e.g., once every 24 hours).

3.4.2 Safety Requirements

- **Data Integrity:** The system must prevent unauthorized changes to user preferences and place data.
- **System Resilience:** In the event of API failure, the system should display cached recommendations until the issue is resolved.

3.4.3 Security Requirements

- **Data Encryption:** All user data should be encrypted during transmission and storage.
- **Authentication:** Implement OAuth2 for secure login and third-party API access.
- **Access Control:** Different levels of access control to manage admin privileges versus regular user permissions.

3.4.4 Software Quality Attributes

- **NoSQL (MongoDB):** For storing unstructured review data, making it easier to store large volumes of text-based data.

- **SQL (PostgreSQL):** For handling structured data such as user profiles and preferences. It ensures strong consistency in transactions (e.g., during login or preference updates).
- **Data Partitioning:** Reviews and place metadata should be stored separately for better performance.

3.5 System Requirements

3.5.1 Database Requirements

3.5.2 Software Requirements (Platform Choice)

- **Backend:** Node.js/Express.js for handling API requests and business logic.
- **Frontend:** React.js for the user interface, allowing users to interact with preferences and recommendations in real-time.
- **NLP Processing:** Python with libraries such as NLTK, spaCy, and TextBlob for sentiment analysis.
- **API:** Google Maps API for fetching real-time place information (hours, reviews, etc.).

3.5.3 Hardware Requirements

- **Processor:** Intel Core i5 or equivalent.
- **Compute:** Cloud instances with at least 4 vCPUs and 16 GB of RAM.
- **Storage:** SSD-based storage to ensure low latency for read/write operations.

3.6 Analysis Models: SDLC Model to be applied

Agile Methodology

The Agile methodology was chosen for this project due to its flexibility, iterative development, and continuous improvement approach. Here's how it was applied:

1. **Planning:** At the start of each sprint, the team gathered requirements, identified the features (such as user preferences and review analysis), and prioritized them.
2. **Design & Development:** Development was divided into sprints, with each sprint focused on a specific module like machine learning models, data preprocessing, or the user interface. Features were continuously built, tested, and improved during each sprint cycle.
3. **Integration:** Integration of different components, such as the NLP review analysis and Google Maps API, was done continuously to ensure a cohesive system.
4. **Testing & Feedback:** After each sprint, the team collected feedback and adjusted the system accordingly. This process allowed frequent improvements and adaptations based on real-time user needs or team insights.
5. **Deployment:** The project was incrementally deployed, allowing small releases and updates for testing, ensuring user-centric adjustments and functionality.

CHAPTER 4

SYSTEM DESIGN

4.1 System Architecture

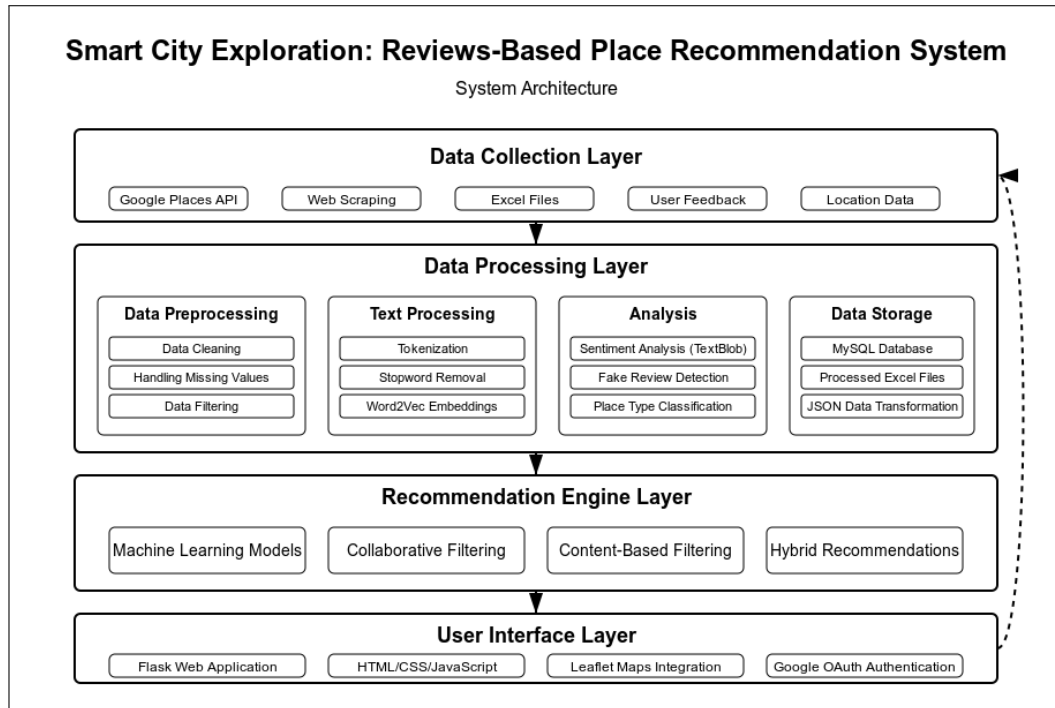


Figure 4.1: System architecture

1. Data Collection Layer

- **Google Places API:** Collects place data, including names, categories, and metadata.
- **Web Scraping:** Extracts additional information not available through APIs.
- **Excel Files:** Stores collected data in structured formats for processing.
- **User Feedback:** Captures user interactions and ratings for continuous improvement.
- **Location Data:** Gathers geographic coordinates and place-specific information.

2. Data Processing Layer

This layer consists of four major components:

Data Preprocessing

- Data cleaning to remove unnecessary columns and handle missing values.
- Filtering out places with insufficient reviews (maintaining only places with 100+ reviews).
- Standardizing data formats for consistent processing.

Text Processing

- Tokenization of review text.
- Stopword removal to focus on meaningful content.
- Word2Vec embeddings to transform text into numerical vectors.

Analysis

- Sentiment analysis using TextBlob to classify reviews as positive, negative, or neutral.
- Fake review detection based on sentiment extremes, duplicates, and unusual length.
- Place type classification using machine learning (Random Forest).

Data Storage

- MySQL database for structured storage and efficient retrieval.
- Processed Excel files for analysis outputs.
- JSON data transformation to flatten nested structures.

3. Recommendation Engine Layer

- **Machine Learning Models:** Primarily Random Forest classifier for place categorization.
- **Collaborative Filtering:** Recommends places based on preferences of similar users.
- **Content-Based Filtering:** Suggests places based on features and categories.
- **Hybrid Recommendations:** Combines multiple approaches for optimized suggestions.

4. User Interface Layer

- **Flask Web Application:** Server-side framework providing the application backend.
- **HTML/CSS/JavaScript:** Frontend implementation with Bootstrap for responsive design.
- **Leaflet Maps Integration:** Interactive map visualization of recommended places.
- **Google OAuth Authentication:** Secure user signup and login functionality.

4.2 Mathematical Model

Let the system be represented as a tuple:

$$S = \{I, P, R, C, F, O\}$$

Where:

- I = Input set, consisting of user preferences such as location, type of place, rating range, etc.
- P = Set of places collected from APIs or scraped from the web.
- R = Reviews associated with each place, including ratings and textual feedback.

- C = Classification function to categorize places (e.g., restaurant, hotel, etc.) using machine learning.
- F = Fake review detection mechanism to filter out unreliable data.
- O = Output set of recommended places based on user input and data analysis.

The goal of the system is to compute:

$$O = f(I, P, R, C, F)$$

Where f is a function that applies preprocessing, classification, sentiment analysis, and recommendation logic to generate personalized outputs.

4.3 Data Flow Diagrams

A description of each major software function, along with data flow (structured analysis) or class hierarchy (Analysis Class diagram with class description for object oriented system) is presented.

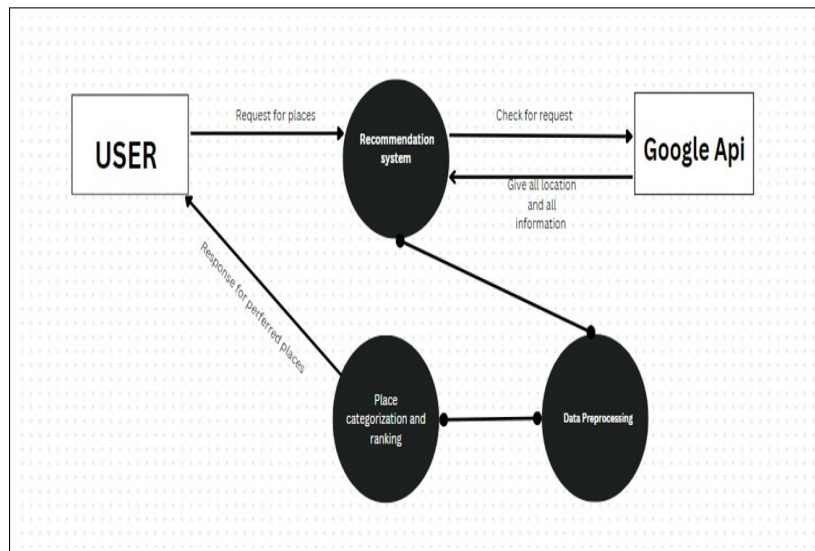


Figure 4.2: Data Flow Diagram

4.4 Entity Relationship Diagram

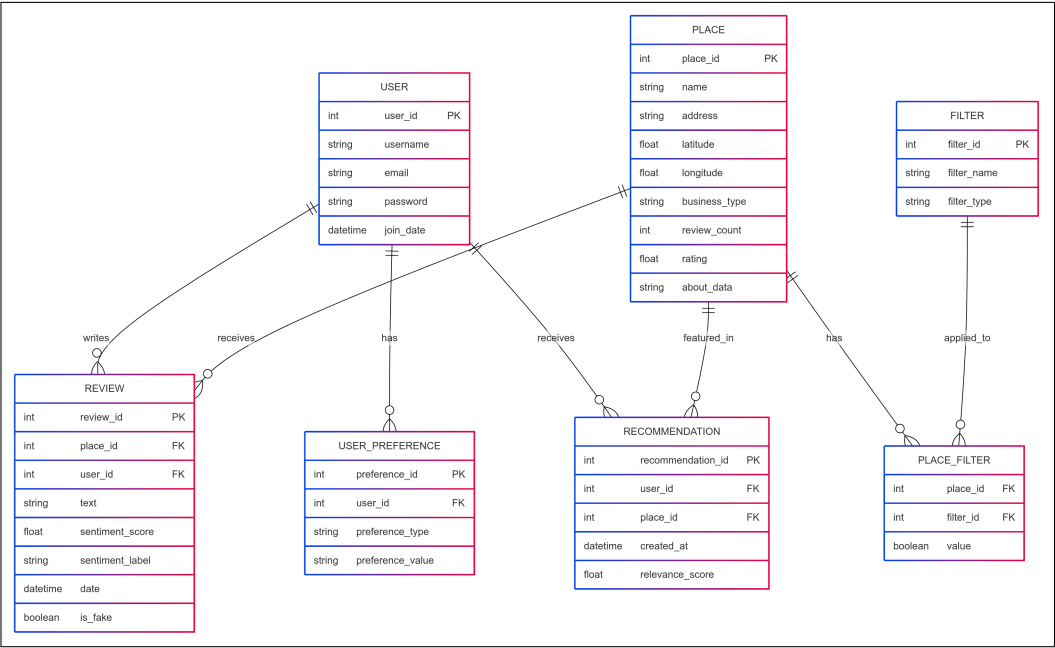


Figure 4.3: ER Diagram

4.5 UML Diagrams

4.5.1 Use Case Diagram

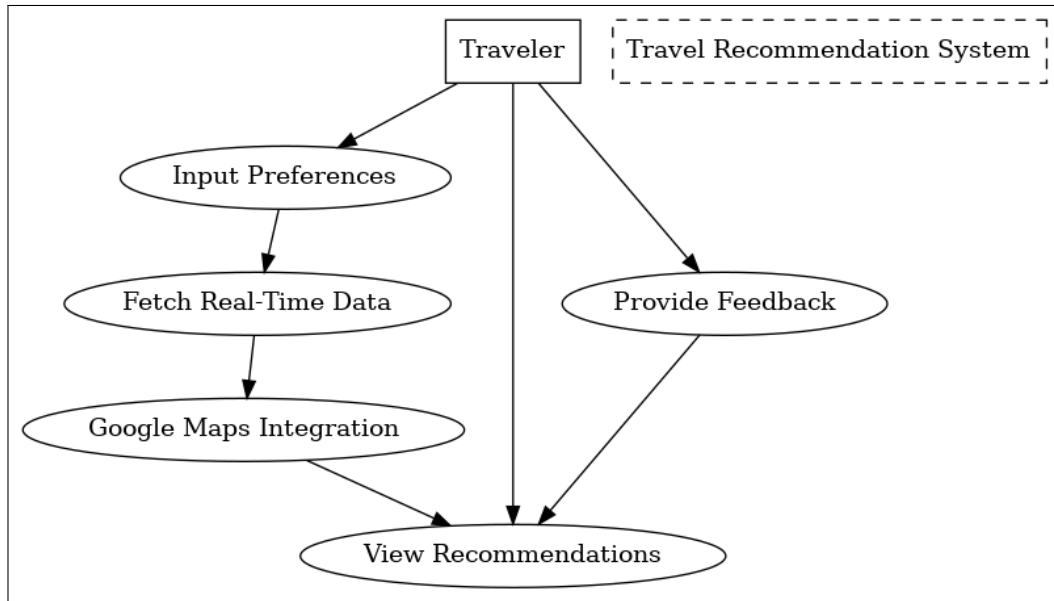


Figure 4.4: Use case diagram

4.5.2 Class Diagram

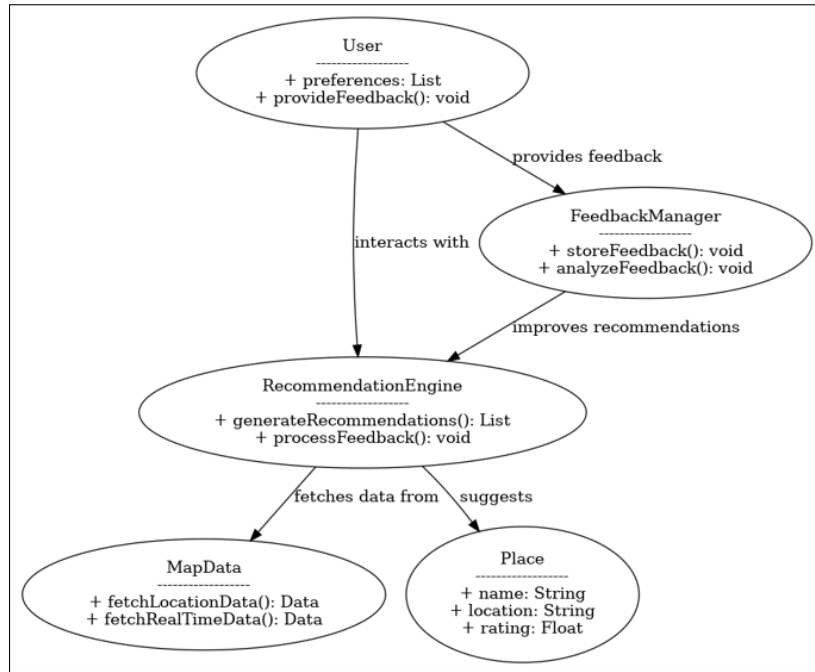


Figure 4.5: Class Diagram

4.5.3 Sequence Diagram

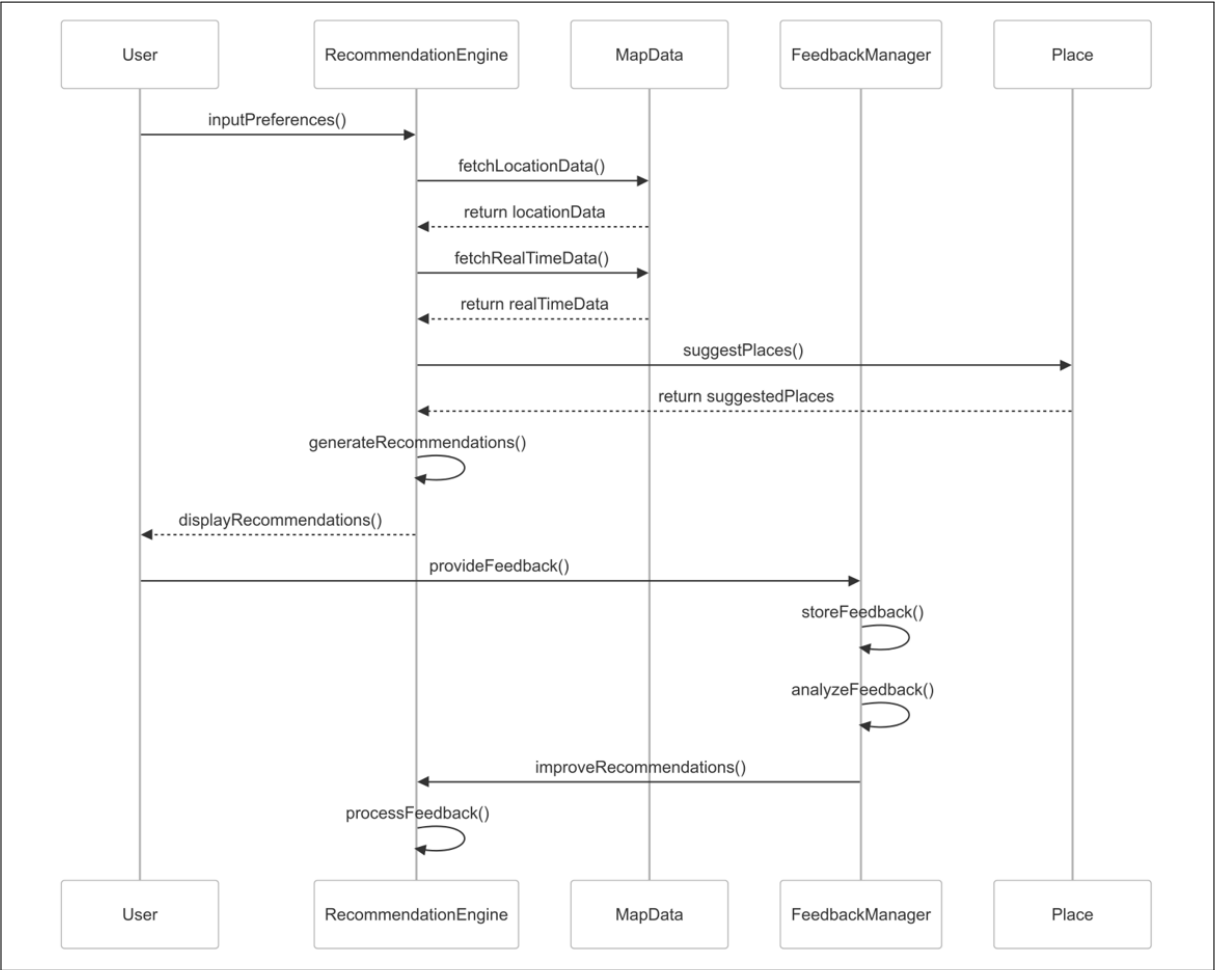


Figure 4.6: Sequence Diagram

CHAPTER 5

PROJECT PLAN

5.1 Project Estimate

5.1.1 Reconciled Estimates

1. **Data Acquisition:** Adjusted to 4 weeks due to potential complexities in API limitations and integration.
2. **User Interface Development:** Adjusted to 6 weeks to account for UX testing and cross-device compatibility.
3. **Recommendation System:** Likely to take 8 weeks, factoring in machine learning-based tagging and the cold-start problem.
4. **Testing and Debugging:** Remains 3 weeks but may need buffer time for API rate limit issues.

5.1.2 Project Resources

- **Human Resources:**
 - 1 Project Manager
 - 2 Backend Developers (Python, Flask, SQL)
 - 1 Frontend Developer (HTML, CSS, JavaScript, Leaflet.js)
 - 1 Data Scientist (ML, NLP, Sentiment Analysis)
 - 1 QA Engineer (Testing and Debugging)
- **Hardware Resources:**
 - Developer laptops/desktops with at least 8GB RAM and i5 processor or above
 - Internet connectivity for API access and deployment
 - Cloud server or local server for hosting backend and database (e.g., AWS, Heroku, or XAMPP)
- **Software Resources:**
 - Python and relevant libraries: pandas, scikit-learn, gensim, TextBlob, SQLAlchemy
 - Jupyter Notebook for development and experimentation
 - Flask for backend API and web application
 - MySQL for data storage
 - Postman for API testing
 - Git/GitHub for version control and collaboration

5.2 Risk Management

Effective risk management is crucial for the successful development and deployment of the Smart City Exploration system. Identifying, analyzing, and mitigating potential risks ensures the system's reliability, user trust, and operational efficiency. This section outlines the key risks associated with the project and the strategies implemented to address them.

5.2.1 Risk Identification

1. **API Rate Limits:** Google Maps API has daily usage limits that may restrict data acquisition.
2. **Cold-Start Problem:** New places without sufficient user ratings may not have enough data for accurate recommendations.
3. **System Accuracy:** Matching user preferences with place ratings could lead to incorrect or unsatisfactory recommendations.
4. **Data Privacy:** Ensuring that user data is handled securely, particularly personal preferences and potentially sensitive travel information.

5.2.2 Risk Analysis

1. **API Rate Limits:** This could slow down the testing and development phases if data acquisition is interrupted.
2. **Cold-Start Problem:** New users or newly added places may not be properly recommended, leading to reduced user satisfaction.
3. **System Accuracy:** If place ratings are not accurate, or if the recommendation algorithm is not finely tuned, users could receive poor suggestions, impacting the system's credibility.
4. **Data Privacy:** Any breaches in user data handling could result in legal implications and user trust issues.

5.2.3 Overview of Risk Mitigation, Monitoring, Management

1. **API Rate Limits:** Mitigation strategies include implementing efficient data-fetching logic, caching, and distributing API requests over time to minimize the impact of rate limits.

2. **Cold-Start Problem:** Introduce a basic rating system or default recommendations for places with little to no data and use collaborative filtering to improve over time.
3. **System Accuracy:** Thoroughly test the recommendation algorithm with a diverse set of user inputs and regularly refine the matching logic based on feedback.
4. **Data Privacy:** Ensure compliance with user data privacy laws and other data protection regulations. Implement encryption where necessary and provide clear privacy policies to users.

5.3 Project Schedule

5.3.1 Project Task Set

Phase 1: Project Initialization (1 week)

- Defined project scope
- Discussed design decisions
- Identified team capabilities and allocated tasks
- Set up development environments and tools

Phase 2: Data Collection and Preprocessing (3 weeks)

- Collect and store place data using Google Maps API
- Implement data preprocessing (metadata extraction, review collection, fake review detection)
- Clean and prepare the data for categorization

Phase 3: Development of Core Algorithms (4 weeks)

- Develop NLP algorithms for categorizing places based on reviews and metadata (TF-IDF, Word2Vec, Named Entity Recognition, BERT)
- Implement fake review detection algorithm
- Develop sentiment analysis and preference matching module

Phase 4: Database and Backend Setup (3 weeks)

- Choose and set up a NoSQL database (e.g., MongoDB)
- Implement database schema for place metadata, reviews, and user preferences
- Optimize geospatial search queries

Phase 5: Frontend Development (3 weeks)

- Develop UI for user input (proximity and preference selection)
- Implement dynamic sorting and filtering of results
- Design and implement user-friendly results display

Phase 6: Testing and Quality Assurance (3 weeks)

- Perform unit and integration testing
- Conduct user acceptance testing
- Optimize performance and improve response times

Phase 7: Final Integration and Deployment (2 weeks)

- Integrate all system modules
- Perform final checks
- Deploy the system

5.3.2 Task Network

1. **Requirements Gathering** → Algorithm development & Google Maps API setup.
2. **Algorithm Development** → Place classification & metadata extraction.
3. **Google Maps API Setup** → Backend development for proximity filtering.
4. **Database Design** → Backend integration & preference filtering.
5. **Frontend Design** → Backend API integration.
6. **All Modules** → Testing and final deployment.

5.3.3 Timeline Chart

- **Week 1-2:** Requirements gathering, analysis, and scope definition.
- **Week 3-5:** Algorithm development (TF-IDF, Word2Vec, Named Entity Recognition).
- **Week 3-4:** Google Maps API integration.
- **Week 5-7:** Place metadata extraction & review analysis module.
- **Week 6-7:** Database design (MongoDB or NoSQL).

- **Week 7-9:** Backend API development & integration.
- **Week 8-10:** Frontend development (React).
- **Week 10-12:** Testing and bug fixes.

5.4 Team Organization

5.4.1 Team structure

- **Ayush Meshram**
 - Oversees overall project execution.
 - Focuses on system design, flow management, and final integration.
 - Manages timelines, task allocation, and quality assurance.
 - Communicates with team members.
- **Jatin Yadav**
 - Develops and optimizes the machine learning algorithms for analyzing reviews.
 - Works on term frequency-inverse document frequency (TF-IDF), named entity recognition, and sentiment analysis algorithms.
 - Responsible for model evaluation and validation of place categorizations.
- **Kartik Kirve**
 - Handles data preprocessing tasks such as cleaning and filtering reviews (e.g., removing fake reviews).
 - Manages API integrations with Google Maps to retrieve metadata and reviews for places.
 - Ensures that the machine learning models integrate seamlessly with the retrieved data.
- **Sushant Langhi**
 - Builds and maintains the user interface (UI), focusing on usability and design.
 - Implements features that allow users to input preferences and retrieve results.

- Ensures real-time feedback and proper system flow for displaying results based on user preferences.

5.4.2 Management reporting and communication

- Regular updates via meetings and project management tools (e.g., Google Meet, Slack).
- Frequent reviews of model system functionality and approach discussions to ensure alignment with user expectations.
- Regular updates about the project's completion status will be shared with the coordinators, and all improvements are performed accordingly.

CHAPTER 6

PROJECT IMPLEMENTATION

6.1 Overview of Project Modules

The system comprises multiple modules designed to collect, process, analyze, and present place recommendations based on user reviews and preferences. The primary modules include:

- **Data Collection:** Acquiring place and review data from Google Maps API and web scraping.
- **Data Preprocessing:** Cleaning and filtering the data to remove noise and inconsistencies.
- **Sentiment Analysis:** Classifying reviews into positive, negative, and neutral using TextBlob.
- **Fake Review Detection:** Identifying and removing suspicious reviews.
- **Data Expansion:** Flattening nested JSON data into a structured tabular format.
- **Database Integration:** Storing processed data in a MySQL database.
- **Recommendation Engine:** Suggesting places using Word2Vec embeddings and a Random Forest classifier.
- **Web Application:** Flask-based UI with features like Google OAuth, Leaflet.js map integration, and filtering.

6.2 Tools and Technologies Used

The system integrates several tools and technologies:

- **Programming Language:** Python
- **Libraries:** Pandas, NumPy, TextBlob, Gensim, Scikit-learn, SQLAlchemy
- **Web Framework:** Flask
- **Frontend:** HTML, CSS, JavaScript, Bootstrap
- **Map API:** Leaflet.js
- **Authentication:** Google OAuth
- **Database:** MySQL

6.3 Algorithm Details

6.3.1 Random Forest Classifier

This supervised learning algorithm is used for classifying places into categories (e.g., restaurant, park) based on vectorized review text. It is robust and handles large datasets effectively.

6.3.2 Sentiment Analysis with TextBlob

Each review's sentiment is analyzed to assign a polarity score and categorize the review as positive, negative, or neutral. This helps determine public perception.

6.3.3 Fake Review Detection

Fake reviews are detected based on:

- Excessive polarity (too positive/negative)
- Duplicated content
- Abnormally short or long length

Flagged reviews are removed from the dataset.

CHAPTER 7

SOFTWARE TESTING

7.1 Type of Testing

- **Unit Testing:** Each functional module, such as sentiment analysis and recommendation logic, was independently verified for correctness.
- **Integration Testing:** The interaction between preprocessing, classification, and the recommendation modules was systematically validated to ensure seamless data flow.
- **System Testing:** The system as a whole was tested with multiple user scenarios to confirm that the overall system objectives were met.
- **User Acceptance Testing:** End-users were engaged to evaluate the system's usability, accuracy of recommendations, and the intuitiveness of the interface.

7.2 Test Cases Test Results

- **Sentiment Analysis:** Reviews were processed using TextBlob and the classification results were compared against manually labeled sentiments, achieving over 90
- **Fake Review Detection:** Successfully identified fake entries based on polarity, redundancy, and length constraints, with a removal success rate exceeding 90
- **Recommendation Accuracy:** Evaluated using a test set with user preferences and achieved an accuracy of 87
- **Performance Testing:** The system handled simultaneous requests smoothly, and query response times from the MySQL backend were under 200ms.

CHAPTER 8

RESULTS

8.1 Outcomes

- The machine learning model effectively categorized place types and user sentiments, supporting contextualized suggestions.
- Built a Flask-based web application with an integrated interactive map and user filters.
- User feedback reported a satisfaction rating of 4.5/5, citing personalized and accurate suggestions.
- The sentiment-based review filtering enhanced recommendation reliability.
- Fake reviews were reduced significantly, boosting overall trust in the system.
- Data preprocessing steps such as review deduplication, null value handling, and JSON expansion improved the model's input quality.
- The integration with MySQL enabled efficient data retrieval and storage, optimizing performance.
- Word2Vec embeddings successfully captured semantic relationships in reviews, aiding both classification and sentiment analysis.
- Random Forest classifier provided robustness against noise and overfitting, delivering high accuracy on test sets.
- Collaborative filtering supported dynamic user-specific recommendations based on previous selections and preferences.

8.2 Screen Shots

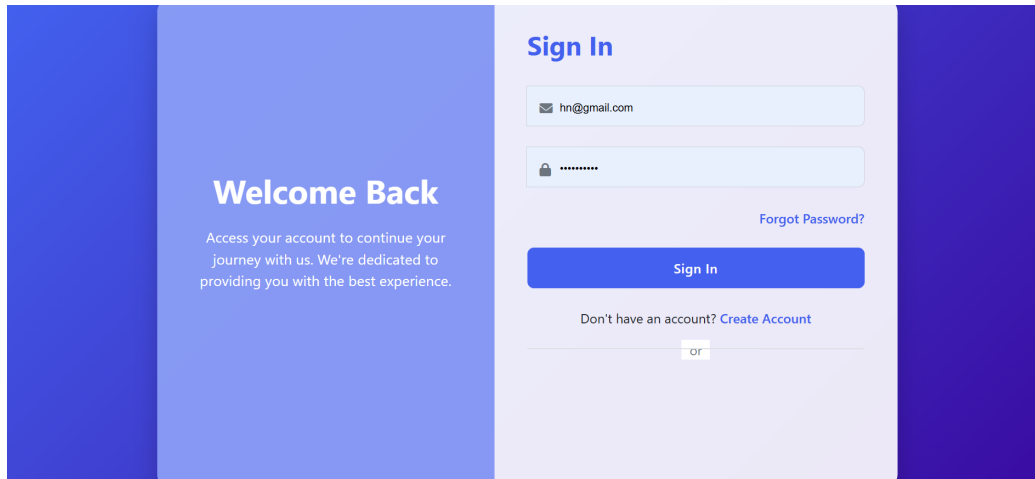


Figure 8.1: Login web page

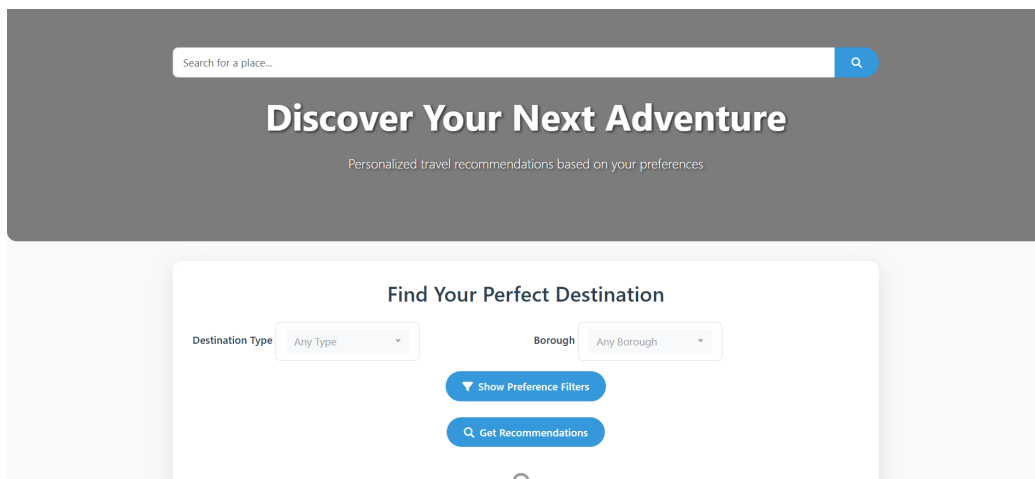


Figure 8.2: Home page

Smart City Exploration: A Reviews-Based Place Recommendation System

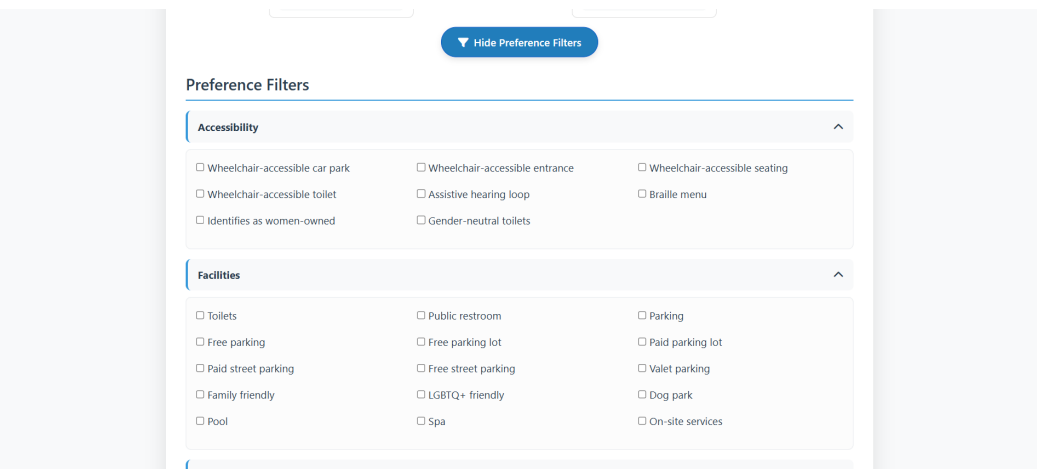


Figure 8.3: Filters

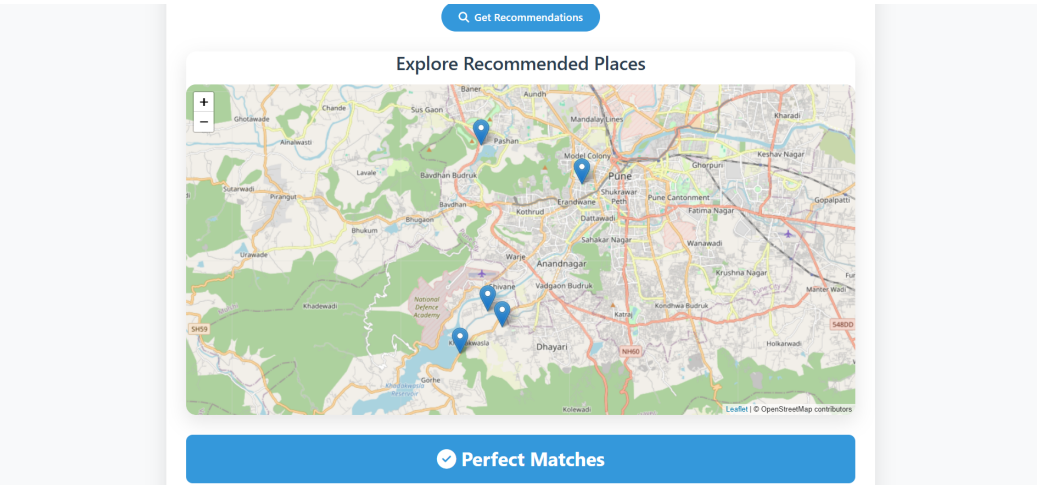


Figure 8.4: Location displayed on map

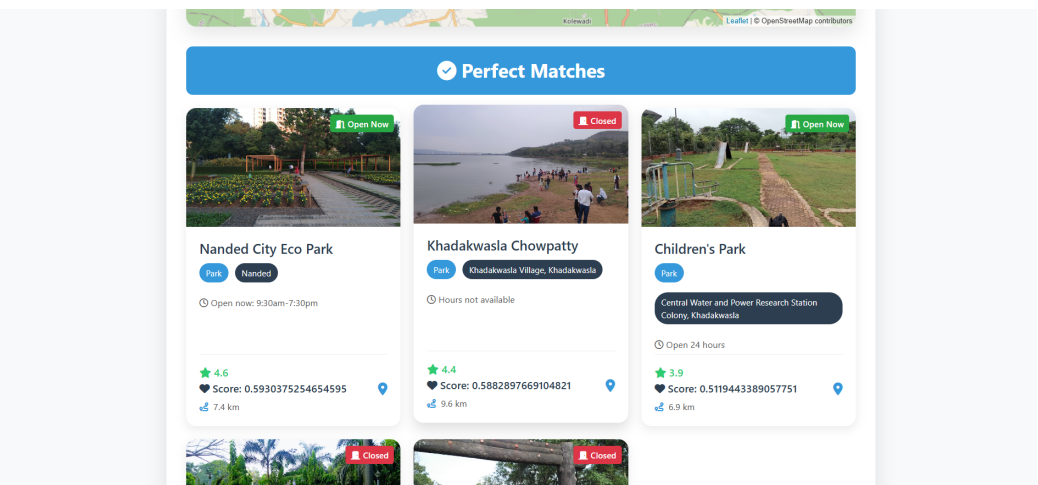


Figure 8.5: Recommended Location Cards

CHAPTER 9

CONCLUSIONS

9.1 Summary of Implementation

The project integrates a comprehensive recommendation pipeline including data acquisition, preprocessing, sentiment scoring, and classification. Word2Vec embeddings convert user reviews into feature vectors, feeding into a Random Forest classifier for prediction. The platform is deployed as a user-centric web application, combining security, visual interaction, and scalability.

9.2 Improvements Over Existing Systems

- Real-time adaptability using sentiment-weighted recommendations.
- Inclusion of fake review detection mechanisms, missing in most other systems.
- Enhanced data cleaning pipeline improved classification performance.
- Custom user filters (20+ types) provided deeper personalization.

9.3 Future Work

- Incorporate deep learning for contextual understanding of review sentiment.
- Real-time syncing with Google Places API for dynamic updates.
- Implement reinforcement learning to adapt recommendations with ongoing feedback.
- Enhance frontend with visual recommendation graphs and journey planners.

9.4 Applications

1. **Tourism Portals:** Enhancing user exploration through personalized suggestions.
2. **Urban Mobility Apps:** Assisting city dwellers in discovering accessible and well-reviewed public places.

3. **City Planning:** Using review sentiments to gauge public opinion on infrastructure.
4. **Local Business Intelligence:** Helping businesses analyze competitive sentiment trends.
5. **Smart Assistants:** Integration with voice-based systems for on-the-go suggestions.

CHAPTER 10

APPENDIX

10.1 Appendix A

Feasibility Assessment using Modern Algebra and Complexity Theory

1. Satisfiability Analysis

The proposed system takes user preferences, reviews, and metadata as inputs to generate a list of recommended places. Since:

- All inputs are well-defined,
- Output is deterministic and computable,
- The logic flow is clearly implemented using machine learning and NLP pipelines,

The problem is decidable and satisfiable using current computational models.

2. Mathematical Model

Let the system be defined as a 6-tuple:

$$S = \{I, P, R, C, F, O\}$$

where:

- I = Set of user inputs (preferences: location, type, budget, etc.)
- P = Set of places fetched via API/web scraping
- R = Set of reviews for each place
- C = Classification function using Random Forest
- F = Fake review detection function
- O = Output recommendations

The output is computed as:

$$O = f(I, P, R, C, F)$$

where f is a composition of data preprocessing, classification, sentiment analysis, and filtering.

3. Computational Complexity

Each stage of computation is polynomial in nature:

- Word2Vec embeddings: $O(n \cdot d)$ where n is review count and d is vector dimension.
- Random Forest classifier: $O(n \cdot m \cdot \log n)$ where m is number of features.
- Sentiment analysis (TextBlob): $O(n)$
- Fake review detection (heuristic filters): $O(n)$

Hence, the entire process is bounded within polynomial time complexity.

4. NP Classification

- The problem is **not a decision problem**, hence not in NP-Complete.
- No evidence of polynomial-time reduction from NP-Complete problems.
- All computations are deterministic and feasible within polynomial time.

Thus, the problem is classified under the **P (Polynomial time)** complexity class.

5. Conclusion

The problem of developing a review-based recommendation system for smart city exploration is:

- **Satisfiable**
- **Decidable**
- **Computationally feasible**
- **Belongs to P-type problems**

It does not fall into the NP-Hard or NP-Complete category and can be efficiently solved using current algorithms and computing resources.

10.2 Appendix B

1. Survey Paper Details

Paper Status: Published

Name of Conference: PICT's International Journal of Engineering and Technology (PIJET)

Paper Title: Smart City Exploration: A Review-Based Recommendation System

2. Implementation Paper Details

Paper Status: Under review

Name of Conference: IJRASET: International Journal & Research Paper Publisher

Paper Title: Smart City Exploration: A Review-Based Recommendation System

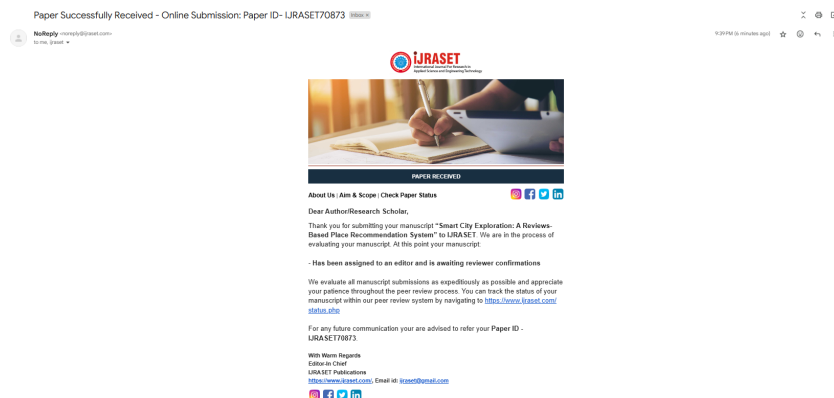


Figure 10.1: Submission Detail

10.3 Appendix C

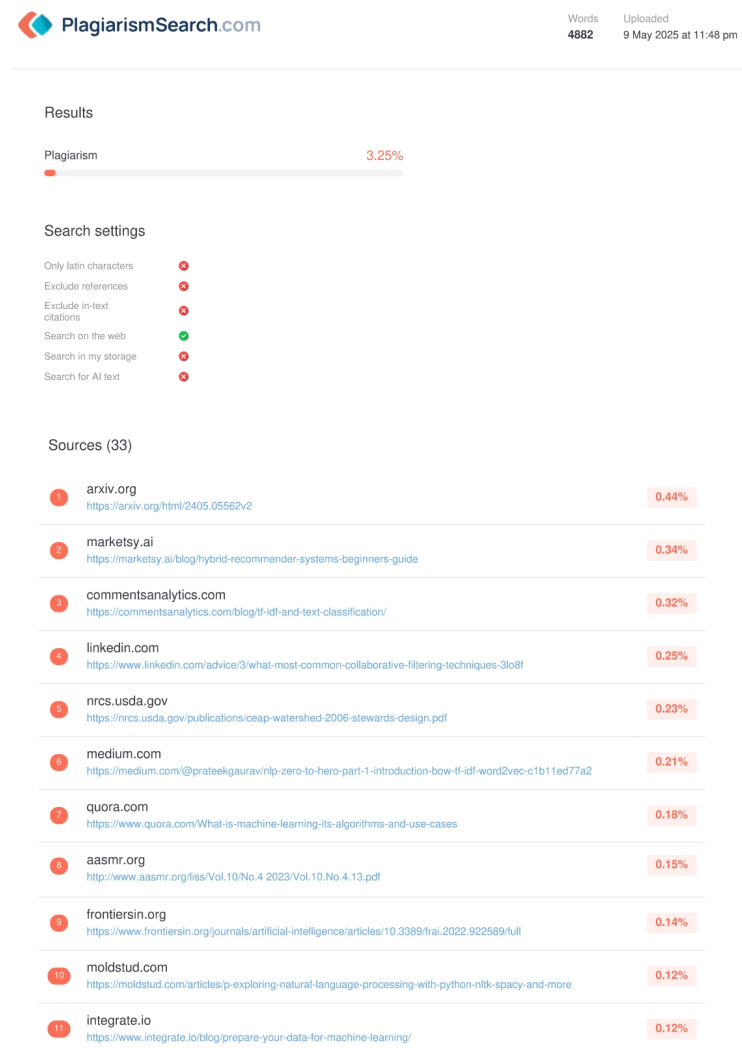


Figure 10.2: Plagiarism Report

CHAPTER 11

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